A SURVEY ON VISION-TO-MUSIC GENERATION: METHODS, DATASETS, EVALUATION, AND CHALLENGES

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ABSTRACT

Vision-to-music generation, including video-to-music and image-to-music tasks, is a significant branch of multimodal artificial intelligence demonstrating vast applications like film scoring and short video creation. However, research in vision-to-music is still in its preliminary stage due to its complex internal structure and the difficulty of modeling dynamic relationships with video. To the best of our knowledge, existing surveys focus on general music generation without comprehensive discussion on vision-to-music. In this paper, we systematically review the research progress in the field of vision-to-music generation. We first analyze the technical characteristics and core challenges for three input types: general videos, human movement videos, and images, as well as two output types of symbolic music and audio music. We then summarize the existing methodologies from the architecture perspective. A detailed review of common datasets and evaluation metrics is provided. Finally, we discuss current challenges and future directions. We hope our survey can inspire further innovation in vision-to-music generation and the broader field of multimodal generation in academic research and industrial applications.

1. INTRODUCTION

Recent advances in multimodal artificial intelligence have witnessed substantial progress in generating and understanding content for modalities like text, images, video, and speech [1–7]. Music generation, as an important part of this multimodal ecosystem, has also seen remarkable development. Among the various music generation tasks (*e.g.* unconditional music generation [8, 9] and text-to-music generation [10, 11]), **vision-to-music**, including **video-to-music** and **image-to-music generation**, has garnered particular interest due to its practical applications in film scoring, short video platforms, and music accompaniment. For general users, automatically generated background music can alleviate copyright concerns and reduce the time spent searching for suitable music. For professional composers, AI-assisted music composition can streamline the iterative process of matching a score to visual content, expediting the communication cycle with directors and producers.

Despite this demand, the development of vision-to-music remains relatively preliminary. For academic research, the inherent challenges ranging from aligning rich visual cues with musical structure to handling the multi-faceted nature of music generation contribute to the task's high complexity, making it more difficult than the common text-to-music task [10–12]. Although a growing number of works have emerged in recent years [13–18], they are still far from meeting the diverse requirements of real-world scenarios. For industrial integration, while other AI-generated content (AIGC) fields, such as text-to-image [19–21] and text-to-video [22,23], have experienced rapid adoption in both professional and consumer contexts, vision-to-music systems have yet to see broad industrial deployment, with only pilot products like Tianpuyue AI [1]. The unique demands of film scoring, which often require precise emotional and temporal synchronization with visual storytelling, heighten the difficulty of achieving robust and artistically consistent results through AI methods.

To the best of our knowledge, although existing works provide reviews on general music generation [24–28], there lack surveys focusing on the vision-to-music generation task. Given the above gaps, we aim to provide a comprehensive survey on vision-to-music generation. We provide a timeline of representative works in Fig. 1, and an overview of vision-to-music generation in Fig. 2.

The subsequent sections of this paper are organized as follows: Sec. 2 introduces the fundamentals of vision-to-music generation, analyzing the technical characteristics and core challenges of three major scenarios: general videos, human movement videos, and images. Sec. 3 reviews current vision-to-music methods, comparing innovations and limitations in vision encoding, vision-music projection, and music generation module design. Sec. 4 discusses recent vision-to-music datasets. Sec. 5 introduces

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[1] https://www.tianpuyue.cn/video2music

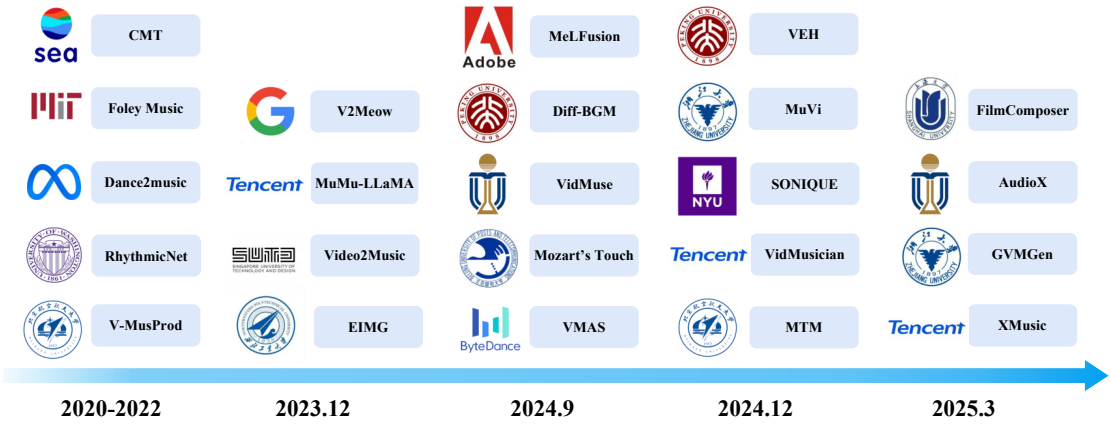


Figure 1: Timeline of representative works in vision-to-music generation.

evaluation metrics, categorized their purposes (music-only and vision-music correspondence) and approaches (objective and subjective). Sec. 6 discusses the current research status and existing challenges. Through this work, we aspire to inspire further innovation in vision-to-music generation and the broader field of multimodal learning communities, driving progress in both academic research and industrial applications of vision-to-music generation.

2. FUNDAMENTALS

In the broad multimodal research community, music is often treated as a subset of audio [8, 22, 29–33]. However, unlike general audio which may include background noise, speech, or sound effects, music embodies intricate internal structures and richness of information, including harmony, counterpoint, and instrumentation. These complexities make it essential to consider music as an independent modality, which sets the stage for exploring vision-to-music generation.

When delving into this specific area, we first need to recognize the unique relationship between visual input and musical output. We analyze the characteristics of three input types in vision-to-music generation: *general videos*, *human movement videos*, and *images*, and two output types: *symbolic music* and *audio music*. This categorization helps us better understand the current state and challenges of the field.

2.1 Input Types

General Videos. This includes a wide range of video contents, such as natural landscapes, films, sports, animations, etc. Techniques in this category typically focus on extracting features like motion, color, or visual semantics to create music that aligns with the visual narrative.

Images. Approaches in this domain focus on transforming static images into music. Since images lack temporal semantics and rhythm, these methods usually only need to focus on the overall style, and there is no strict requirement for the duration of the generated music. The application scenarios for image-to-music are not as extensive as video-to-music, but they include functionalities like creating musical memories for photo albums. This type of pair data is easy to collect, but its inherent correlation may not be very strong.

Human Movement Videos. These videos typically include dance, sports, instrument performances, and other human movements. For instrument performance videos, where humans play music instruments but the audio is removed, the music is determined by the input video to some extent, and the generation process is similar to reconstructing music from the silent videos. For dance, sports, and other human movements, they emphasize rhythmic alignment (especially local rhythm) more than general videos, while semantic constraints are generally weaker, requiring only overall style matching. Therefore, extracted 2D/3D keypoints representing human motion are often directly used as inputs instead of raw videos.

In the remaining sections, we will mainly focus on general videos and images, while paying relatively less attention to human movement videos. This is because they focus on rhythmic relations and the semantic association with music is relatively weak, where 2D body keypoints are directly used as video features. Their application scenarios are also relatively limited [13].

2.2 Output Types

Symbolic Music. Symbolic music is represented as discrete elements like notes, chords, or sequences of musical symbols [25]. Most early vision-to-music methods are symbolic [13, 15, 34, 35]. Symbolic music can incorporate music theory, such as chords, and generate longer pieces with good controllability. However, the limited data availability restricts its scalability to large models, and the expressive and emotional depth is constrained by soundfonts.

Audio Music. Such methods aim to generate music in its audio form [14, 16–18, 36–39], often employing generative models such as transformers [40], VAEs, GANs [41], or diffusion models [42] to synthesize realistic sound from the visual input. Audio music benefits from large-scale datasets for training, enabling end-to-end generation with rich expressiveness and performance. However, despite efforts of adding controls in audio generation [43], the controllability of audio music is still relatively weak, com-

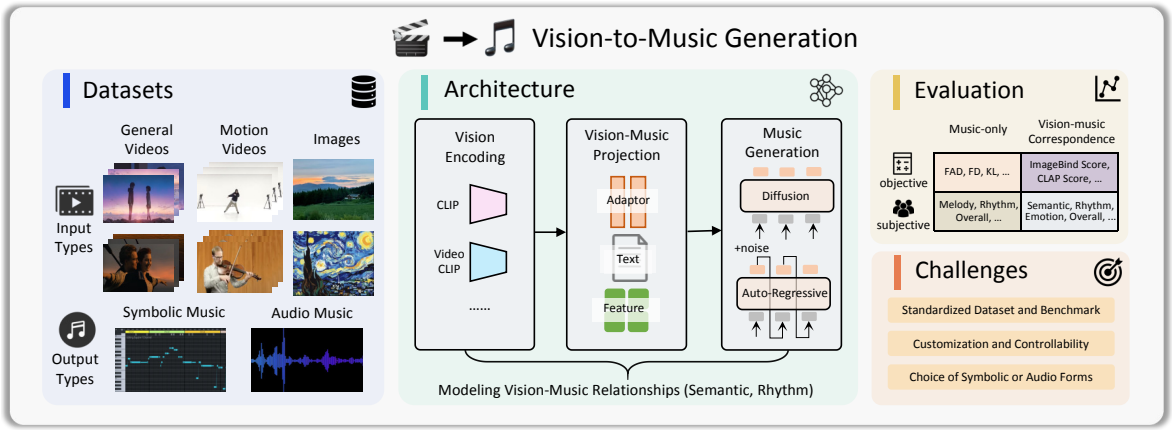


Figure 2: Overview of vision-to-music generation.

pared to symbolic music where multiple control signals can be used, *e.g.* pitch, duration, instrument, rhythm, chord, etc. Moreover, the generated music is typically shorter due to sampling rate limitations, *e.g.* usually under 20 seconds for methods in Tab. 1, where symbolic methods can easily achieve whole-song length.

3. METHODS

In this section, we discuss the existing works on vision-to-music generation. We summarize vision-to-music generation methods in Tab. 1.

3.1 Tasks

We begin by categorizing the methods based on the input types outlined in Sec. 2.

General Video-to-Music. Early general video-to-music methods were usually symbolic [13, 15, 34, 35]. With the development of audio-form music generation [11, 12], a large number of audio-based general video-to-music works have emerged in the past few years [16–18, 36–39, 44–46].

Image-to-Music. Early image-to-music works were also primarily symbolic [47–50], where models analyze color, texture, and semantic content to generate music. Recent works [16, 38, 51, 52] generated audio-form music from multiple modalities (video, image, and text).

Human Movement Video-to-music. For dance or sports videos, existing methods focus on extracting rhythmic patterns from dance videos and mapping them to musical rhythm generation [31, 53–59]. For music performance videos, current methods learn to reconstruct the original music from the silent videos [60–63].

3.2 Architecture

The architecture of vision-to-music systems can be broken down into three major components: vision encoding, vision-music projection, and music generation.

Vision Encoding. This stage is focused on extracting features from the input video or image. A commonly used vision encoder is CLIP [66], which is pretrained on massive image-text pairs to achieve open-domain visual understanding capabilities. Video understanding backbones [64, 70, 78, 85, 88, 92, 98] are also used to extract spatiotemporal features. Some also use additional encoders for color information [15], emotion information [46], or intermediate text features [37]. For human movement videos, it is important to extract motion features for rhythmic alignment, *e.g.*, directly calculating first-order difference from human keypoints as motion velocity [31, 53], or using pre-trained motion encoder [14, 54].

Vision-Music Projection. This component involves mapping the visual features into the music space. Most methods directly use the visual features as the input of the music generation model, or through simple cross-attention mechanisms [15, 35, 37, 52]. Some methods design specialized adapters [16, 17, 36] for better feature alignment, *e.g.* to capture temporal-related or local features. Besides using feature-based mapping, some studies suggest using text as an intermediate representation of the visual features [18, 38, 44, 87] and subsequently utilizing text-to-music models for music generation. Some symbolic music generation methods use symbolic elements as the vision-music mapping [13, 99].

Music Generation. Once the visual and music features have been aligned, the next step is to generate the musical output. This stage can be tackled using auto-regressive or diffusion-based generative models. Auto-regressive models [40] can be used for both symbolic [13, 15, 99] and audio music generation [17, 18, 37, 39, 44, 45]. Diffusion models [42, 86] can be used for symbolic [35] music to directly generate piano rolls, but mostly for audio music [12, 38, 46, 101].

3.3 Vision-Music Relationships

Vision-music relationships establish the correspondence between videos and music. Unlike the vision-music projection discussed in the previous section (which focuses on the architecture), the relationships discussed here focus on the overall correspondence between input and output. These relationships can be broadly classified into two categories: semantic relationships and rhythmic relationships.

Semantic Relationships. This type of relationship focuses on how visual elements (such as color, objects, or scenes)

Table 1: Methods for vision-to-music generation. Sem: Semantics. Rhy: Rhythm. AR: Auto-regressive. Diff.: Diffusion.

Method	Demo	Date	Input Type	Modality	Music Length	Vision-Music Relationships	Vision Encoding	Vision-Music Projection	Music Generation
▼ *General Videos and Images:*									
CMT [13]	Link	2021/11	General Video	Symbolic	3min	Rhy	-	Elements	AR (CP [9])
V-MusProd [15]	Link	2022/11	General Video	Symbolic	6min	Sem, Rhy	CLIP2Video [64], Histogan [65]	Feature	AR (CP [9])
V2Meow [14]	Link	2023/05	General Video	Audio	10sec	Sem, Rhy	CLIP [66], I3D Flow [67], ViT-VQGAN [68]	Feature	AR
MuMu-LLaMA [51] (M^2UGen [16])	Link	2023/11	General Video, Image	Audio	30sec	Sem	ViT [69], ViViT [70]	Adapter	AR (LLaMA2 [71])
Video2Music [34]	Link	2023/11	General Video	Symbolic	5min	Sem, Rhy	CLIP [66]	Feature	AR
EIMG [72]	Link	2023/12	Image	Symbolic	15sec	Sem	ALAE [73], β-VAE [74], VQ-VAE [75]	Adapter	VAE (FNT [76], LSR [77])
Diff-BGM [35]	Link	2024/05	General Video	Symbolic	5min	Sem	VideoCLIP [78]	Feature	Diff. (Polyffusion [79])
Mozart's Touch [44]	Link	2024/05	General Video, Image	Audio	10sec	Sem	BLIP [80]	Text	AR (MusicGen [11])
MeLFusion [52]	Link	2024/06	Image	Audio	10sec	Sem	DDIM [81] + T2I LDM [82]	Feature	Diff.
VidMuse [17]	Link	2024/06	General Video	Audio	20sec	Sem	CLIP [66]	Adapter	AR (MusicGen [11])
S2L2-V2M [83]	Link	2024/08	General Video	Audio	10sec	Sem	Enhanced Video Mamba	Adapter	AR (LLaMA2 [71])
VMAS [45]	Link	2024/09	General Video	Audio	10sec	Sem, Rhy	Hiera [84]	Feature	AR
MuVi [36]	Link	2024/10	General Video	Audio	20sec	Sem, Rhy	VideoMAE V2 [85]	Adapter	Diff. (DiT [86])
SONIQUE [87]	Link	2024/10	General Video	Audio	20sec	Sem, Rhy	Video-LLaMA [88], CLAP [89]	Text	Diff. (Stable Audio [90])
VEH [91]	-	2024/10	General Video	Symbolic	30sec	Sem	VideoChat [92]	Text	AR (T5 [93])
M2M-Gen [94]	Link	2024/10	Image (Manga)	Audio	1min	Sem	CLIP [66], GPT-4 [1]	Text	AR (MusicLM [95])
HPM [46]	Link	2024/11	General Video	Audio	10sec	Sem	CLIP [66], TAVAR [96], WECL [97]	Feature	Diff. (AudioLDM [29])
VidMusician [37]	Link	2024/12	General Video	Audio	30sec	Sem, Rhy	CLIP [66], T5 [93]	Adapter	AR (MusicGen [11])
MTM [38]	Link	2024/12	General Video, Image	Audio	30sec	Sem	InternVL2 [98]	Text	Diff. (Stable Audio Open [12])
XMusic [99]	Link	2025/01	General Video, Image	Symbolic	20sec	Sem, Rhy	ResNet [100], CLIP [66]	Elements	AR (CP [9])
GVMGen [39]	Link	2025/01	General Video	Audio	15sec	Sem	CLIP [66]	Adapter	AR (MusicGen [11])
AudioX [101]	Link	2025/03	General Video	Audio	10sec	Sem	CLIP [66]	Feature	Diff. (Stable Audio Open [12])
FilmComposer [18]	Link	2025/03	General Video	Audio	15sec	Sem, Rhy	Controllable Rhythm Transformer, GPT-4v [1], Motion Detector	Text	AR (MusicGen [11])
▼ *Human Movement Videos:*									
Audeo [61]	Link	2020/06	Performance Video	Symbolic	30sec	Rhy	ResNet [100]	Feature	GAN
Foley Music [62]	Link	2020/07	Performance Video	Symbolic	10sec	Rhy	2D Body Keypoints	Feature	AR
Multi-Instrucment Net [60]	-	2020/12	Performance Video	Audio	10sec	Rhy	2D Body Keypoints	Feature	VAE
RhythmicNet [53]	Link	2021/06	Dance Video	Symbolic	10sec	Rhy	2D Body Keypoints	Feature	AR (REMI [102])
Dance2Music [57]	Link	2021/07	Dance Video	Symbolic	12sec	Rhy	2D Body Keypoints	Feature	AR
D2M-GAN [54]	Link	2022/04	Dance Video	Audio	2sec	Rhy	2D Body Keypoints, I3D [103]	Feature	GAN
CDCD [55]	Link	2022/06	Dance Video	Audio	2sec	Rhy	2D Body Keypoints, I3D [103]	Feature	Diff.
LORIS [31]	Link	2023/05	Movement Video	Audio	50sec	Rhy	2D Body Keypoints, I3D [103]	Feature	Diff.
VisBeatNet [56]	-	2024/01	Dance Video	Symbolic	Realtime	Rhy	2D Body Keypoints	Feature	AR
UniMuMo [104]	Link	2024/10	Dance Video	Audio	10sec	Rhy	2D Body Keypoints	Feature	Diff.

relate to musical components (such as mood, melody, or chords). For music performance videos, the music is determined by the video instead of a general and implicit semantic relationship [61, 62]. For dance and movement videos, the semantics in the video is not utilized. Symbolic methods [15, 34, 99] explicitly define semantic, color, and emotion relationships extracted from pretrained models to utilize the controllability of symbolic music. Recent audio-based methods generally use a single vision encoder to extract semantic features. These semantic features are usually global and insensitive to semantic changes within the video. Some methods [17, 36, 39] also design special modules to enhance local semantic correspondence. However, for most audio methods generating 10-second music, the concept of "local" may not have a significant impact.

Rhythmic Relationships. Rhythmic relationships mainly refer to the correspondence between the rhythm of the video (*e.g.* local movements, scene transitions, global video rhythm) and the rhythm of the music (*e.g.* local beats, global tempo). For human movement videos, such as dance or instrument playing, rhythmic relationships become significant, especially the correspondence between local rhythm and human movements. For general videos, early works [13–15, 45] use optical flow or RGB Difference to represent the video rhythm. Recent works mostly do not consider rhythm information or use frame-by-frame semantic features to implicitly provide local rhythm correspondence [17, 37], which is not prominent in the generated music. In methods that use text for vision-music projection [38, 87], the video content is used to generate requirements for musical rhythm, such as the rhythm of each scene or the overall tempo.

4. DATASETS

In this section, we introduce common datasets for vision-to-music. Plenty of datasets have been proposed in the literature of the vision-to-music field, and different methods often use different datasets for training and testing. Therefore, it is necessary to organize and analyze these datasets. Common datasets are listed in Tab. 2.

4.1 Input Categories

Based on the types of videos/images in vision-music datasets, we categorize the datasets as follows:

General Videos. Videos in these datasets are usually sourced from platforms like YouTube. Most datasets focus on Music Videos [15, 34, 35, 45, 105, 107], as they have satisfactory video-music alignment and are easier to collect. Other datasets include a variety of video types, such as trailers, advertisements, animations, and documentaries [17, 37], or subsets from larger datasets like AudioSet [108]. These videos offer better diversity, but the video-music alignment may be weaker, requiring strict fil-

Table 2: Datasets for vision-to-music generation.

Dataset	Access	Date	Source	Modality	Size	Total Length (hour)	Avg. Length (second)	Annotations
▼ *General Videos:*								
HIMV-200K [105]	Link	2017/04	Music Video (Youtube-8M [106])	Audio	200K	-	-	-
MVED [107]	Link	2020/09	Music Video	Audio	1.9K	16.5	30	Emotion
SymMV [15]	Link	2022/11	Music Video	MIDI, Audio	1.1K	76.5	241	Lyrics, Genre, Chord, Melody, Tonality, Beat
MV100K [14]	-	2023/05	Music Video (Youtube-8M [106])	Audio	110K	5000	163	Genre
MusicCaps [95]	Link	2023/01	Diverse Videos (AudioSet [108])	Audio	5.5K	15.3	10	Genre, Caption, Emotion, Tempo, Instrument, Rhythm, ...
EmoMV [109]	Link	2023/03	Music Video (MVED [107], AudioSet [108])	Audio	6K	44.3	27	Emotion
MUVideo [16]	Link	2023/11	Diverse Videos (Balanced-AudioSet [108])	Audio	14.5K	40.3	10	Instructions
MuVi-Sync [34]	Link	2023/11	Music Video	MIDI, Audio	784	-	-	Scene Offset, Emotion, Motion, Semantic, Chord, Key, Loudness, Density, ...
BGM909 [35]	Link	2024/05	Music Video	MIDI	909	-	-	Caption, Style, Chord, Melody, Beat, Shot
V2M [17]	-	2024/06	Diverse Videos	Audio	360K	18000	180	Genre
DISCO-MV [45]	-	2024/09	Music Video (DISCO-10M [110])	Audio	2200K	47000	77	Genre
FilmScoreDB [46]	-	2024/11	Film Video	Audio	32K	90.3	10	Movie Title
DVMSet [37]	-	2024/12	Diverse Videos	Audio	3.8K	-	-	-
HarmonySet [111]	Link	2025/03	Diverse Videos	Audio	48K	458.8	32	Description
MusicPro-7k [18]	Link	2025/03	Film Video	Audio	7K	-	-	Description, Melody, Rhythm Spots
▼ *Human Movement Videos*								
URMP [112]	Link	2016/12	Performance Video	MIDI, Audio	44	1.3	106	Instruments
MUSIC [113]	Link	2018/04	Performance Video	Audio	685	45.7	239	Instruments
AIST++ [114]	Link	2021/01	Dance Video (AIST [115])	Audio	1.4K	5.2	13	3D Motion
TikTok Dance-Music [54]	Link	2022/04	Dance Video	Audio	445	1.5	12	-
LORIS [31]	Link	2023/05	Dance Video, Sports Video (AIST [115], FisV [116], FS1000 [117])	Audio	16K	86.43	19	2D Pose
▼ *Images*								
Music-Image [118]	Link	2016/07	Image (Music Video)	Audio	22.6K	377	60	Lyrics
Shuttersong [119]	Link	2017/08	Image (Shuttersong App)	Audio	586	-	-	Lyrics
IMAC [120]	Link	2019/04	Image (FI [121])	Audio	3.8K	63.3	60	Emotion
MUImage [16]	Link	2023/11	Image (Balanced-AudioSet [108])	Audio	14.5k	40.3	10	Instructions
EIMG [72]	Link	2023/12	Image (IAPS [122], NAPS [123])	MIDI	3K	12.5	15	VA Value
MeLBench [52]	Link	2024/06	Image (Diverse Videos)	Audio	11.2K	31.2	10	Genre, Caption

tering [16,95]. FilmScoreDB and MusicPro-7k [18,46] focus on film scores, where the music has a deeper semantic correspondence with the video and serves as an accompaniment rather than being the primary focus, as in music videos. Recently, some datasets also provide textual descriptions of videos and music [38, 111, 124] to assist text-bridged video-to-music generation methods. 3. **Human Movement Videos.** These videos can be divided into instrument performances and dance/sport categories. Instrument performance datasets [112, 113] aim to reconstruct music from instrumental performance videos. Dance/sport datasets [31,54,114] focus on generating music from dance or sports videos, emphasizing local rhythmic alignment while downplaying semantic relationships.

Images. Existing image-to-music datasets are relatively scarce. Sources of the images are usually frames from music videos [52,118] or existing image datasets [16,72,120].

4.2 Music Domains

Vision-music datasets can be divided into MIDI and audio based on the music modality. MIDI datasets [15, 34, 35] are created by transcribing audio into the MIDI format or sourced from existing music-only datasets [125]. Audio datasets contain only raw audio files.

Compared to audio datasets, MIDI datasets have the following advantages: (1) More annotations like Chord, Melody, Beat, Tonality, etc; (2) Longer average duration enables generating longer music pieces; (3) Suitable for training both symbolic and audio music generation models. However, a significant limitation of MIDI datasets is their smaller scale (*e.g.* 1K songs, 100 hours vs. 100K-2M songs, 5K-50K hours) and relatively limited diversity.

5. EVALUATION

Common metrics for vision-to-music are categorized in Tab. 3 and 4. The evaluation of vision-to-music generation can be divided into two categories: **objective** and **subjective**. Objective evaluation uses fixed rule-based algorithms or existing models to extract features and calculate musical metrics. It is relatively objective and convenient for fair comparison, but has certain biases and cannot cover all aspects of music generation, often differing significantly from human subjective perception. Similar to other generation tasks [126, 127], subjective evaluation is typically used in vision-to-music generation for a more comprehensive assessment, *i.e.* conducting user studies where participants rate/compare music generated by different models.

From another perspective, metrics can be divided into **music-only** and **vision-music correspondence** based on assessment purposes. The former only evaluates whether the music itself is pleasant/realistic/structurally complete, etc., while the latter focuses on the correspondence between the music and the visual input.

For **music-only objective metrics**, symbolic music generation methods [13, 15, 35, 72, 83, 99] use some statistics-based methods to calculate certain pitch or rhythm-related statistical metrics of MIDI, such as Scale Consistency, Pitch Entropy, etc. These metrics are usually compared with ground truth music, and the closer they are, the more realistic the music is considered. Audio music

Table 3: Objective metrics of vision-to-music generation. M: MIDI. A: Audio. V: Video. I: Image. T:Text. Pit: Pit. Rhy: Rhythm. Fid: Fidelity. Sem: Semantic.

Metric	Used in Paper	Input	Type
▼ *Music-only:*			
Scale Consistency	[15, 83]	M	Pit
Pit Entropy	[15, 72, 83]	M	Pit
Pit Class Histogram Entropy	[13, 15, 35, 83, 99]	M	Pit
Empty Beat Rate	[15, 83, 99]	M	Rhy
Average Inter-Onset Interval	[15, 83]	M	Rhy
Grooving Pattern Similarity	[13, 35, 99]	M	Rhy
Structure Indicator	[13, 35]	M	Rhy
Frechet Audio Distance (FAD)	[14, 16–18, 36, 39, 52] [44–46, 83, 91, 101]	A	Fid
Frechet Distance (FD)	[14, 17, 36–38, 52, 87, 101]	A	Fid
Kullback-Leibler Divergence (KL)	[14, 16–18, 36–39, 44–46] [52, 83, 87, 91, 101]	A	Fid
Beats Coverage Score (BCS)	[36, 46]	A	Rhy
Beats Hit Score (BHS)	[36, 46]	A	Rhy
Inception Score (IS)	[36, 46, 101]	A	Fid
▼ *Vision-music Correspondence:*			
ImageBind Score/Rank	[16–18, 37, 38, 44, 83, 101]	A,V/I	Sem
CLAP Score	[37, 87, 91]	A,A/T	Sem
Video-Music CLIP Precision	[15, 83]	A,V	Sem
Video-Music Correspondence	[35]	A,V	Sem
Cross-modal Relevance	[39]	A,V	Sem
Temporal Alignment	[39]	A,V	Rhy
Rhythm Alignment	[37]	A,V	Rhy

generation methods widely adopt metrics such as Frechet Audio Distance (FAD), Frechet Distance (FD) [2] , and Kullback Leibler Divergence (KL) to evaluate the similarity between generated music and ground truth music. Some methods [36, 46] also introduce metrics like BCS and BHS to measure rhythmic similarity based on music beats.

Objective metrics for vision-music correspondence usually focus on the audio modality. The most commonly used are ImageBind Score/Rank and CLAP score, which leverage pretrained multimodal models like ImageBind [131] and CLAP [89] for similarity evaluation. Some methods [15, 35, 39, 83] have also designed specific vision-music retrieval evaluation metrics, with slight differences in model selection and retrieval methods. Additionally, GVMGen [39] and VidMusician [37] have designed objective metrics to evaluate the rhythmic correspondence between visions and music. However, since the pretrained models are usually trained with general audio data instead of specified music data, these objective metrics commonly do not perfectly align with human judgments.

Subjective metrics mainly include MOS (generally using a 5-point Likert scale), pair preference (*i.e.* win rate), and ranking different music. Common subjective metrics in vision-to-music generation are given in Tab. 4. The selection of specific subjective metrics depends on the vision-music relationship emphasized by the method.

6. CHALLENGES

Despite advances in vision-to-music generation, we identify several key challenges for the academic community:

Lack of Standardized Objective Datasets and Benchmarks: The training and evaluation datasets differ across models, sometimes leading to comparisons between models fine-tuned on proprietary datasets and those evaluated via zero-shot inference on other datasets. This disparity significantly undermines the fairness of model comparisons and makes it challenging to identify the state-of-the-art. Besides, current evaluation metrics often do not align with actual human perception, *e.g.* FAD and KL are based on general audio data rather than on music-specific data, and symbolic metrics are statistically based and exhibit low correlation with human preferences. Though most papers provide demos for qualitative comparisons, they are prone to issues such as cherry-picked examples, insufficient sample size, and subjectivity in evaluating the outputs.

Limited Customization and Controllability: Most existing models function as black boxes, making it challenging to personalize or control attributes of the generated music, such as style, instrumentation, and rhythm. This significantly affects the models' practical applicability.

Trade-off Between Symbolic and Audio Forms: As discussed in Sec. 2, audio-based methods benefit from large-scale data but generally offer limited controllability and are constrained by the computational cost of high-fidelity generation, often resulting in shorter musical pieces. In contrast, symbolic approaches, while limited by available data, offer better controllability and can produce longer compositions. A promising direction is to combine symbolic and audio methods to achieve a better trade-off.

Beyond academic challenges, exploring how to align these technologies with music industry applications (such as film and game scoring) and end-user products (like short-video platform background music) in consumer products offers significant commercial opportunities.

Table 4: Subjective metrics of vision-to-music generation.

Metric	Used in Paper
▼ *Music-only:*	
Music Melody	[15, 35]
Music Rhythm	[15, 35]
Music Richness	[39, 99]
Audio Quality	[17, 36]
Overall Music Quality	[13, 14, 17, 18, 34, 38, 39, 44, 45, 52, 91, 94]
▼ *Vision-music Correspondence:*	
Semantic Consistency	[15, 18, 35–38]
Rhythm Consistency	[15, 18, 34, 35, 38, 91, 99]
Emotion Consistency	[38, 91, 99]
Overall Correspondence	[13–18, 34, 39, 44, 45, 52, 83, 87, 94]

7. CONCLUSION

In this paper, we review the recent advancements in vision-to-music generation, covering both symbolic and audio-based approaches. We identified key technical challenges in visual feature extraction, cross-modal projection, and music generation, and discussed the limitations in current datasets and evaluation metrics. We believe that addressing these challenges will pave the way for future research and applications on vision-to-music generation.

[2] The difference between FAD and FD is the feature extractor: FAD [128] uses VGGish [129], while FD uses PANNs [130].

8. ETHICAL STATEMENT

This work is conducted with a clear awareness of the ethical challenges associated with vision-to-music generation technologies. As models learn to translate visual inputs—such as images, videos, or artworks—into music, they operate at the intersection of multiple cultural, legal, and emotional domains. One major concern is copyright infringement, particularly when models are trained on copyrighted music that closely resemble existing compositions without clear attribution or licensing. Given the creative and expressive nature of music, even stylistic mimicry may raise legal and ethical questions.

In addition, these models risk amplifying cultural or stylistic biases present in the training data. For example, models trained primarily on western classical or pop music may marginalize non-western musical traditions or underrepresent diverse emotional and cultural expressions. The alignment between visual input and musical output can further reinforce problematic stereotypes or fail to capture culturally appropriate interpretations, especially when visual content carries religious or political significance.

A third dimension involves emotional appropriateness. Music is a powerful emotional medium. Generating music from sensitive or traumatic visual content—such as scenes of violence, grief, or historical trauma—may result in emotionally discordant or insensitive outcomes.

To mitigate these risks, we advocate for transparent data collection practices, inclusion of diverse musical and visual cultures, and evaluation frameworks that assess not only perceptual quality but also cultural and emotional alignment. We further encourage interdisciplinary collaboration between technologists, musicians, ethicists, and legal scholars to ensure the responsible development and deployment of these systems.

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